

RSMC — Practical Test Protocol and Assessment (math-clean)

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Bottom line

UNPROVEN. Not useful for practice until effects pass the constrained-null tests below on archived data.

Scope

This document specifies a minimal, falsifiable protocol to test the claimed septimal 7->14 echo, the period-3 sidebands, the Z9 clock with 63-step return, and the cross-domain constant R_{∞} . All math is inline ASCII.

Definitions

- Gates: indexed by $g = 0..G-1$. Gate rate f_g (Hz). Gate period $T_g = 1/f_g$.
- Analytic signals: $A_7(g)$, $A_{14}(g)$ are complex envelopes extracted at the “7” and “14” components by a fixed, preregistered method.
- Phases: $\phi_7(g) = \arg(A_7(g))$, $\phi_{14}(g) = \arg(A_{14}(g))$.
- Closure phase: $\delta\phi(g) = \text{wrap}(2*\phi_7(g) - \phi_{14}(g))$ in $(-\pi, \pi]$.
- Trinity label: $b(g)$ in $\{0,1,2\}$ from a fixed, preregistered classifier on the slow envelope.
- Z9 clock: $\tau_9(g) = t(g) + 3*(b(g) \bmod 3)$, where $t(g) = g$ (integer gate counter).
- Z-series: $z_k = \exp(i2\pi*\tau_9(k)/9)$.
- Bicoherence: let $X(f)$ be the FFT of the rail; $b(f_1, f_2) = |E[X(f_1) * X(f_2) * \text{conj}(X(f_1+f_2))]| / \sqrt{E[|X(f_1) * X(f_2)|^2] * E[|X(f_1+f_2)|^2]}$.
- R_{∞} (provisional): common slope C from shelf-colored echo regression (see Test 1), reported with 95% CI. Cross-domain stability of C is the working definition of the “constant.”

Claimed effects (status: UNPROVEN)

1) Septimal 7->14 quadratic echo with a closure-phase plateau at gates. 2) Period-3 envelope with sidebands at $f_g \pm f_g/3$. 3) Z9 clock peaks at 9 and super-return at 63 (LCM(7,9)). 4) Cross-domain constant R_{∞} under “full holonomy.”

Falsifiers (targeted kill-controls)

- Orientation randomization: randomize per-gate orientations while preserving timing -> shelves and $\pm f_g/3$ must die.
- Gate-order constrained shuffle: permute gates within local blocks to destroy long-range ordering -> 63 peak must die.
- Phase scramble: randomize phases of the raw signal while preserving amplitude spectrum -> bicoherence must die.

- Mis-tune: test 6.8/13.6 and 7.2/14.4 bands -> septimal effects must weaken.
- Seam de-tilt: remove any seam/alignment parameter -> concurrency should collapse.

Data requirements

Archive per rail: - Raw signal or frames; sampling rate; exact start time. - Gate indices {g}, gate boundaries in samples. - Extracted $A_7(g)$, $A_{14}(g)$; $\phi_7(g)$, $\phi_{14}(g)$; $\Delta\phi(g)$. - $b(g)$, $t(g)$, $\tau_9(g)$; z_k . - FFT parameters; windowing; all hyperparameters. - Code and seeds to reproduce all figures and nulls.

Pre-registration

- Freeze detection thresholds, filters, and classifiers for A_7 , A_{14} , and $b(g)$.
- Fix all parameters before seeing outcomes: window sizes, block sizes for shuffles, frequency bins, regression form, significance levels, multiple-comparison correction.
- Primary alpha = 0.01 after correction.

Core tests

Test 1 — Shelf-colored echo regression

Goal: three parallel strata with common slope C . 1) Compute $x_i(g) = \text{unwrap}(\phi_{14}(g) - 2\phi_7(g))$. 2) Partition gates into three “shelves” S_j by a preregistered rule tied to $b(g)$ or seam position; S_1, S_2, S_3 non-overlapping. 3) For each j in $\{1, 2, 3\}$, fit $x_i(g) = a_j + Ct(g) + e(g)$ for g in S_j . 4) Report $\hat{C}$ and $\hat{a}_j$ with 95% CI per stratum. Test H_0 : slopes equal across strata and $C \neq 0$. 5) Run orientation randomization; refit. Require: effect dies under randomization.

Pass criteria: - Slopes across strata agree: $\max_j |\hat{C}_j - \hat{C}_k| \leq \text{tol}_C$. - Global test rejects H_0 : $C = 0$ at $p < 0.01$ (corrected). - Randomization null shows no significant C ($p \geq 0.2$ recommended) and no parallelism.

Test 2 — Z9 clock with 63 super-return

Goal: significant periodogram peaks at 9 and 63 that vanish under constrained shuffles. 1) Form $z_k = \exp(i2\pi\tau_9(k)/9)$ for $k = 0..G-1$. 2) Compute periodogram $P_z(n)$ over n in $1..N$ ($N \geq 128$, preregistered). Focus on $n = 9$ and $n = 63$. 3) Build a gate-order constrained shuffle null: divide the rail into blocks of B gates; permute gates within each block; recompute P_z . Repeat S times.

Pass criteria: - Observed $P_z(9)$ and $P_z(63)$ exceed the 99th percentile of the null across S shuffles. - Under full gate randomization, both peaks drop below the 95th percentile.

Test 3 — Bicoherence 7,7->14

Goal: elevated $b(f_1, f_2)$ near $(f_1, f_2) = (f_7, f_7)$ with $f_{14} = f_7 + f_7$. 1) Compute bicoherence on the raw rail with preregistered windows. 2) Extract summary statistic $B_{\text{star}} = \max_{\{f_1, f_2\} \in N_7} b(f_1, f_2)$, where N_7 is a fixed neighborhood around f_7 . 3) Build two nulls: (a) phase scramble of the rail; (b) mis-tune bands to 6.8/13.6 and 7.2/14.4 and recompute.

Pass criteria: - B_star exceeds 99th percentile of both nulls. - Mis-tune reduces B_star below the 95th percentile of the tuned distribution.

Test 4 — Sidebands at $f_g \pm f_g/3$

1) Compute the spectrum around f_g . 2) Define S_{minus} = power at $f_g - f_g/3$; S_{plus} = power at $f_g + f_g/3$.
3) Compare to orientation-randomization null.

Pass criteria: - Both S_{minus} and S_{plus} exceed 99th percentile of the null.

Certificate (per rail)

Fields to report: - Rail_ID - G (gate count) - f_g (Hz) - Test1_C (slope), CI_low, CI_high, Parallelism_ok (Y/N), Rand_Kill_ok (Y/N), p_Echo - Test2_P9_obs, P63_obs, P9_pnull, P63_pnull, Constrained_Kill_ok (Y/N) - Test3_Bstar_obs, Bstar_p_phase, Bstar_p_mistune - Test4_Sminus_obs, Sminus_p, Splus_obs, Splus_p - Flags: C1,C2,C3 (parallel shelves present Y/N) - Decision: PASS if all required effects significant and all targeted kills succeed; else FAIL

Example certificate line (ASCII table)

```
Rail_ID | G | f_g | C | CI_low | CI_high | Parallelism_ok | Rand_Kill_ok |
p_Echo | P9 | P63 | p9_pnull | p63_pnull | B* | p_phase | p_mistune | S- | p(S-) |
S+ | p(S+) | C1 | C2 | C3 | DECISION
R001    | 620 | 3.00 | 0.012 | 0.009 | 0.015 | Y | Y | 0.004 | 12.3 | 8.7 |
0.006 | 0.009 | 0.41 | 0.004 | 0.007 | 0.22 | 0.004 | 0.19 | 0.006 | Y | Y | Y |
PASS
```

Reporting standard

- Release raw data, code, parameters, and seeds.
- Publish preregistration and analysis scripts.
- For each figure, include a single-file script that regenerates it.

Reviewer mini-figure (one rail)

Panel A: rose of delta_phi with Rayleigh p.

Panel B: periodogram of z_k with marks at 9 and 63; overlay null percentiles.

Panel C: bicoherence heatmap near (f_7, f_7) with null violin for B_star.

Limitations and risks

- Gate detection, orientation extraction, and b(g) labeling can induce artifacts if tuned post hoc. Freeze them.
- Low SNR or drifting f_g will smear sidebands and weaken 63.
- Cross-domain “constant” likely fails without tight geometry control.

TRL and timeline

- TRL: 2-3.
- Single-domain preregistered run: weeks. Two domains with independent replication: 3-6 months.

Decision rule

- If any targeted kill control fails to kill the corresponding effect, invalidate the effect.
- If any required effect does not meet $p < 0.01$ (corrected), FAIL.
- Claim R_{∞} only if two independent domains yield overlapping 95% CI for C.

Appendix — Pseudocode

A1. Orientation randomization

```
for repeat in 1..S:  
  permute orientation angles across gates; keep gate times fixed  
  recompute A_7, A_14, phi_7, phi_14, delta_phi  
  run Tests 1 and 4; store stats
```

A2. Gate-order constrained shuffle

```
B = preregistered block size (e.g., 16 gates)  
for repeat in 1..S:  
  for each consecutive block of B gates:  
    permute gates within the block  
  recompute tau_9, z_k, periodogram P_z; store P_z(9), P_z(63)
```

A3. Phase scramble

```
X = FFT(raw)  
randomize phases: X' = |X| * exp(i*theta), theta ~ Uniform(-pi,pi)  
rail' = IFFT(X')  
recompute bicoherence; get B_star
```

A4. Mis-tune control

```
repeat band extraction at 6.8/13.6 and 7.2/14.4  
recompute bicoherence and echo regression  
compare to tuned results
```

A5. Sideband check

```
compute spectrum near f_g
S_minus = power at f_g - f_g/3
S_plus  = power at f_g + f_g/3
compare to orientation-randomization null percentiles
```

A6. Echo regression fit

```
xi = unwrap(phi_14 - 2*phi_7)
for j in {1,2,3}:
    select S_j by preregistered rule
    fit xi = a_j + C*t + e via OLS or circular-linear if using delta_phi
    test equality of slopes; report C_hat and CI
```

End